How soon is now?

Tackling Africa's recurrent aflatoxin crises through integrated, multi-sectoral management strategies

December 2024

Alejandro Ortega-Beltran, Ranajit Bandyopadhyay, and Simeon Ehui
International Institute of Tropical Agriculture (IITA), Ibadan, Nigeria

Summary

Aflatoxin contamination of staple crops remains a pressing threat to public health, food security and safety, and economic stability in many African countries. The poorest populations are most at risk. Opportunities for trade and investment are severely limited due to the pervasive presence of aflatoxins.

As the harmful effects of aflatoxins continue to impact producers, consumers, and societies, the need for immediate and coordinated action has never been more crucial. Despite ongoing efforts, responses have been fragmented and inconsistent, falling short of providing sustainable solutions to mitigate the aflatoxin challenge. Addressing this crisis requires a coordinated, converged, multi-sectoral approach involving farmers, governments, international organizations, the private sector, and research institutions.

Implementing integrated management strategies that are gender sensitive and socially inclusive is essential to effectively combat aflatoxin contamination across local, regional, and continental levels, thereby improving health outcomes, connecting smallholder farmers to organized markets, and increasing income.

In this policy brief, we emphasize the urgency of dealing with the aflatoxin crises through the integrated approach of Food Convergence Innovation (FCI), which combine biocontrol solutions, such as Aflasafe™, improved agricultural practices, post-harvest technologies, and robust policy interventions backed by awareness, effective marketing linkages, digital tools, and traceability systems. To address the far-reaching consequences of aflatoxin contamination, now is the time for united action to protect public health, enhance food safety, and boost economic development across Africa.

Aflatoxin is a highly toxic substance produced by Aspergillus fungi that contaminates many food crops, including maize, groundnut, sorghum, and chilies.

IITA is a not-for-profit international research center that generates agricultural innovations to meet Africa's pressing challenges of hunger, malnutrition, poverty, and natural resource degradation. Working with partners across Africa, we improve livelihoods, enhance food and nutrition security, increase employment, and preserve natural resource integrity. IITA is a member of CGIAR, a global agricultural research partnership for a food-secure future.

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Background – what are aflatoxins and their impacts

Aflatoxins are highly toxic compounds produced by certain strains of *Aspergillus* fungi, particularly *A. flavus* and *A. parasiticus.* These toxins, regulated at very low levels (parts per billion, ppb), affect several staple and cash crops in many countries in Africa, including maize, groundnut, sorghum, and chilies. In tropical and subtropical regions, favorable climate conditions promote aflatoxin production. In addition, climate change is worsening the aflatoxin challenge, disproportionately affecting low- and middle-income countries, while in high-income countries the negative effects are somehow mitigated because of reliable infrastructure, testing capacity, and strict regulations.

The health impacts of aflatoxins are alarming, particularly in Africa. Chronic exposure is linked to various types of cancer, stunting, immune system suppression, and other long-term problems, while acute poisoning has led to fatal outbreaks, especially in East Africa. The African Union's Partnership for Aflatoxin Control in Africa (PACA) estimates of annual health costs are staggering: US$ 1.6 billion for Nigeria, US$ 1.1 billion for Tanzania, and US$ 577 million for Uganda! Beyond human health, aflatoxins also frequently affect livestock, fisheries, and pets with the most recent reports (2024) being from South Africa, DR Congo, Zambia, and Malawi.

Aflatoxins severely impact Africa's agricultural trade. Crops exceeding aflatoxin tolerance thresholds cannot enter premium local, regional, or international markets. Countries like Uganda and Tanzania have faced bans on maize exports from Kenya, Burundi, and South Sudan, disrupting local economies, disturbing regional harmony, and driving up poverty in rural areas. Companies whose products are banned suffer reputational damage and financial losses. Local food, feed, and brewing industries often reject locally-produced crops and have no other option but to import grains. Recurrent contamination continues to reduce Africa's ability to benefit from trade opportunities, causing hundreds of millions of dollars in losses annually and stalling economic growth.

The situation calls for stronger regulatory frameworks, increased monitoring, and climate-adaptive strategies to minimize the pervasive impact of aflatoxins across the continent.

Farmers applying Aflasafe on a groundnut field during a training event.

What solutions are available?

Addressing aflatoxin contamination through effective management strategies is critical for improving farmer livelihoods and ensuring food security and safety across Africa. Various management strategies have been developed, including improved pre- and post-harvest agricultural practices, tolerant varieties (although not always effective), post-harvest handling (e.g., drying and storage), and biocontrol technologies (e.g., Aflasafe).

A carefully selected combination of aflatoxin management tools, tailored to specific situations, can achieve more cost-effective aflatoxin reduction throughout the value chain.

Benefits in using Aflasafe

- Large-scale trials and commercial usage in multiple countries demonstrate significant reductions in aflatoxin contamination, improving food and feed safety.
- Safe crops produced and utilized for both domestic consumption and export, improving farmer income, increasing availability of safe food, and boosting national economies.
- Availability of locally produced aflatoxin standard-compliant crops lowers import of grains.
- Lower aflatoxin levels reduce the risk of aflatoxin-related health issues, particularly in children who are vulnerable to stunting and immune system suppression.
- Lower aflatoxin levels across the value chain are obtained if biocontrol is combined with any other available management tool.
- Private sector companies in various countries in Africa have invested >US$ 10 million to mass manufacture and distribute the products.

IITA's Ranajit Bandyophadyay (leftmost, front) with Bill Gates (center, in cap) and Gates Foundation staff Jeff Raikes, Sam Dryden, and Prem Warrior, who visited IITA-Kano in 2011. This crucial visit led to the transition of Aflasafe as a laboratory based technology to its industrialization through strategic partnerships, effective commercialization strategies, and sustained investment.

Biocontrol as a centerpiece of integrated management strategies

An effective intervention is the use of Aflasafe, a bioprotectant developed by IITA in partnership with the United States Department of Agriculture – Agricultural Research Service (USDA-ARS), and several national and international partners. Aflasafe products contain non-toxin-producing (atoxigenic) strains of *A. flavus* native to the target countries. These strains outcompete toxin-producing strains, effectively reducing aflatoxin contamination in crops by 80% to 100% compared to non-treated crops. Currently, products are commercially available for use in 12 countries (see map). More products are being developed for another 11 countries.

What policy efforts made by governments and/or economic blocks can be capitalized?

Due to the importance of aflatoxins, African Union Commission created Partnership for Aflatoxin Control in Africa (PACA) to address the aflatoxin problem in Africa through a comprehensive set of policies and programs. Also, policies to prevent contamination and exposure have been launched by Regional Economic Communities [East African Community (EAC) and Economic Community of West African States (ECOWAS)], and individual Governments (e.g., Ghana). Frameworks to prevent contamination and exposure exist but need to be implemented at scale to reduce contamination and exposure on the ground.

Map of status of Aflasafe activities in 23 countries in Africa.

Multi-pronged approach for effective adoption of integrated aflatoxin management strategies

Awareness and education

Educating farmers, extension agents, consumers, industries, and policymakers on aflatoxin risks and integrated management strategies is essential. Training should focus on equipping farmers with the skills to use control technologies effectively, while awareness campaigns should highlight mitigation techniques and the economic benefits of aflatoxin-safe crops. Both efforts must be gender-sensitive and socially inclusive.

Monitoring and evaluation

Establishing systems to track aflatoxin levels in crops before and after use of integrated management strategies is essential. This will provide valuable data on the strategy's impact and enable adjustments to the deployment strategies. Leveraging mobile platforms or digital tools can further enhance real-time reporting and decision-making.

Public-private partnerships

Scaling integrated management strategies effectively requires collaboration between farmer organizations, governments, development agencies, financial institutions, and the private sector. By aligning their efforts, these groups can ensure consistent production, distribution, and adoption of best practices by farmers, aggregators, processors, traders, and other key stakeholders. [See point on Food Convergence Innovation (FCI) on page 1.]

Breaking siloed paradigms

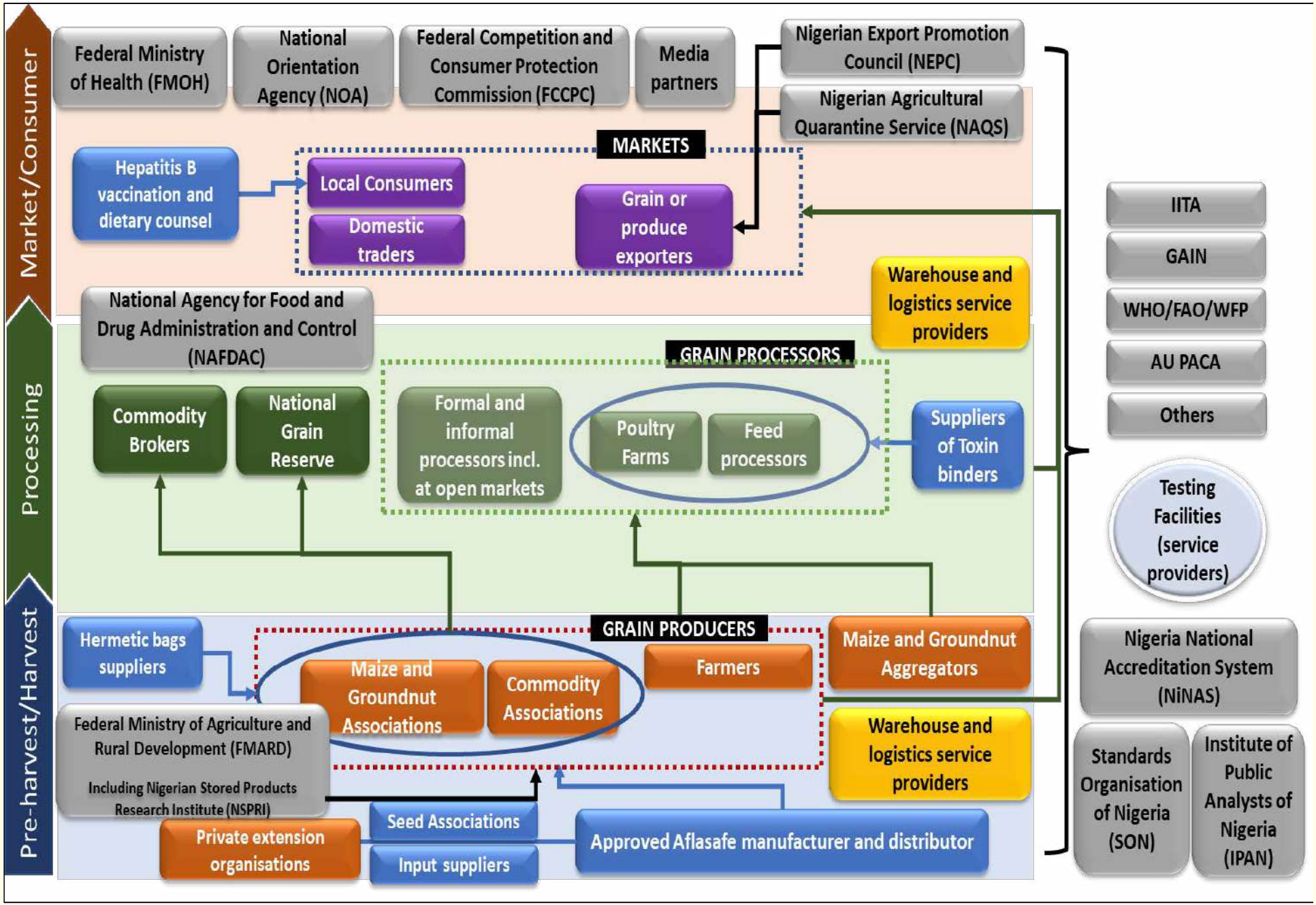
Adopt an FCI approach, which offers a comprehensive solution by integrating data, technology, and multi-sector collaboration. The FCI framework builds on a holistic, digital-driven approach to tackle food safety concerns, including aflatoxin, by creating an ecosystem where stakeholders across the value chain have access to reliable, actionable data, and sustainable solutions. Two aflatoxin/Aflasafe uses cases are at the core of two E.U.-funded international consortia led by IITA (FS4Africa and FCI4Africa).
https://foodsafety4africa.eu/ https://fci4africa.eu/

Appropriate subsidies

Governments should consider providing subsidies or financial incentives to smallholder farmers adopting aflatoxin control strategies. They should also extend support to companies producing such technologies and those paying incentives or premium prices to farmers and aggregators that supply aflatoxin-safe crops. Such measures would increase accessibility and adoption, especially among resource-poor farmers who face financial barriers to adoption.

Research and development

Continuous research and development actions on region-specific atoxigenic isolates of *Aspergillus flavus* and the adaptation of Aflasafe to different agroecological zones not yet reached are essential.



The integrated Food Convergence Innovation (FCI) approach, which combines biocontrol solutions, such as Aflasafe™, improved agricultural practices, post-harvest technologies, and robust policy interventions backed by institutional support, effective market linkages, awareness, digital tools, and traceability systems.

Aflasafe manufacturing facilities constructed by HarvestField Industries Limited in Nigeria (A, B), BAMTAARE SA in Senegal (C), A to Z Textiles Limited in Tanzania (D), and Kenya Agricultural and Livestock Research Organization (KALRO) in Kenya (E).

Policy recommendations

1

Encourage countries that have not developed aflatoxin control policies or adopted the existing policies of Regional Economic Communities to do so.

2

Promote integrated aflatoxin management strategies at national and regional levels. National and State Governments affected by aflatoxin should formally promote integrated aflatoxin control strategies and agricultural policies with gender and socially inclusive lenses, while also establishing regulatory frameworks that support their appropriate implementation.

3

Make food safety an integral component of any food security program implemented in a country. Technologies aimed at increasing yields should be accompanied with measures to empower farmers to produce high-quality, safe food to nourish themselves and consumers.

4

Establish or strengthen national aflatoxin control programs. Countries should establish or strengthen national programs focused on aflatoxin control, which should include setting maximum allowable limits for aflatoxin in food and feed products, establishing deployable aflatoxin testing platforms, conducting regular testing, enforcing compliance with food safety standards, and providing options for repurposing contaminated crops.

5

Facilitate access to aflatoxin-conscious markets. Governments and trade organizations should collaborate to enable smallholder farmers producing aflatoxin-safe crops gaining access to premium local, regional, and international markets. Certification schemes and traceability systems are critical for achieving this goal, along with correct training of farmers in the use of aflatoxin control technologies to ensure compliance with safety standards and maximize market opportunities.

6

Strengthen partnerships. Collaboration among farmer organizations, governments, development partners, international organizations, and research and development institutes such as IITA and other CGIAR centers will be key to ensuring that integrated aflatoxin management strategies are adopted and scaled in a sustainable way. The FCI model can facilitate this by integrating a digital backbone, behavioral insights, and strategic advisory support.

Conclusion

Addressing the aflatoxin challenge is critical for improving food security, health outcomes, and economic stability in Africa. The adoption of integrated management strategies provides a sustainable and effective strategy for reducing aflatoxin contamination in crops.

To ensure widespread impact, farmer organizations, governments, international organizations, the private sector, and research institutions must make decisive and concerted efforts to vigorously promote adoption of these strategies, build capacity, and develop supportive policies.

By integrating these strategies, Africa can reduce the health risks associated with aflatoxins while also contributing to broader development goals, leading to achieve at least 55 targets across 13 of the 17 Sustainable Development Goals (SDGs) in countries where the Aflasafe Initiative operates.

Further information

Ortega-Beltran A, Bandyopadhyay R. 2021. Contributions of integrated aflatoxin management strategies to achieve the Sustainable Development Goals in various African countries. Global Food Security 30: 100559.

Ortega-Beltran A, Bandyopadhyay R. 2023. Aflatoxin biocontrol in practice requires a multidisciplinary, long-term approach. Frontiers in Sustainable Food Systems 7: 1110964.

This work was conducted as part of the CGIAR Initiative on Plant Health, now integrated into the CGIAR Sustainable Farming Science Program.

Contact details

Alejandro Ortega-Beltran, IITA, PMB 5320, Ibadan, Nigeria a.beltran@cgiar.org +234 803 978 4220